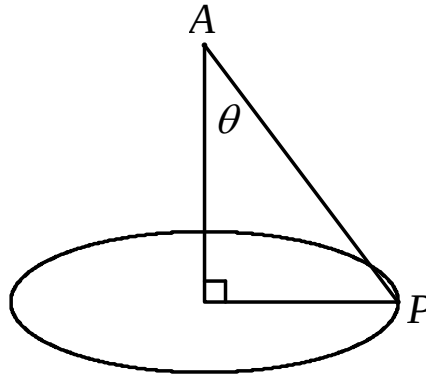


Year 12 Term 2 Mathematics Extension 2, 2009

Question 1

(a) Graph the rectangular hyperbola $xy = 25$ clearly showing the equations of the asymptotes and directrices, and the location of the foci.

(b) From a fixed point A , a particle P of mass 10 grams is suspended by a light inextensible string of length 17 cm as shown.



The particle P rotates with angular velocity ω radians per second in uniform circular motion radius 8cm in the horizontal plane, and has a semi-vertical angle θ with point A .

By resolving the forces on the particle P find the angular velocity ω (to 2 significant figures) when the acceleration due to gravity is 10 m/s^2 .

(c) A point which is initially at rest at the origin, moves to the right with velocity $v \text{ m/s}$.
The displacement $x \text{ metres}$ is given by :

$$x = v - \tan^{-1} v$$

(i) Find $\ddot{x}$ in terms of v .

(ii) Find an expression for the velocity v in terms of time t .

(d) The region bounded by $y=0$, $x = \frac{\pi}{4}$ and the curve $y = \tan^{-1} x$ is rotated about the y axis.

(i) Using the method of cylindrical shells show that the volume of revolution V is given by :

$$V = 2\pi \int_0^{\frac{\pi}{4}} x \tan^{-1} x \, dx$$

(ii) Hence find the volume of revolution V .

Question 2

From the ground a body of unit mass is projected vertically through a resistive medium.

The resistive force in Newtons is $\frac{1}{20}$ of the speed of the body.

The acceleration due to gravity is 10 m/s^2 and the initial velocity is 120 m/s .

(a) For the upward motion :

(i) Draw a diagram showing all the forces acting on the body.

(ii) Show that :

$$\ddot{x} = -10 - \frac{v}{20} \text{ where } v \text{ is the velocity of the body.}$$

(iii) Show that the maximum height reached is ≈ 520 metres .

(iv) Show that the time T seconds to reach the maximum height is ≈ 9.4 seconds.

(b) The body now falls for T seconds.

(i) Draw a diagram showing all the forces acting on the body.

(ii) Find the terminal velocity.

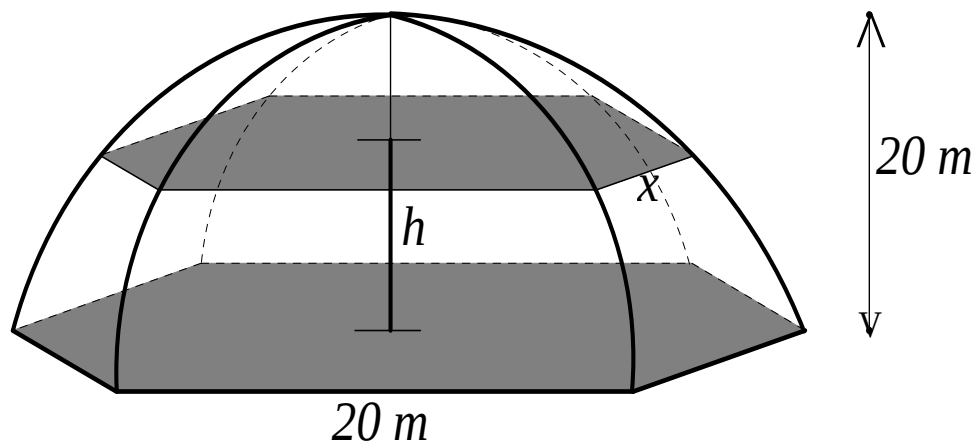
(iii) Find the velocity v (to 2 significant figures) of the body at T seconds.

(iv) Find the distance (to 3 significant figures) the body is above the ground after T seconds.

Question 3

(a) Asymmetrical dome shown has a height of 20 metres and a regular hexagonal base of side 20 metres. The apex of the dome is directly above the centre of the base.

Each strut is a quarter of a circle starting from the corner of the base to of the apex of the dome.



(i) For the slice h metres above and parallel to the base, show that the length x of each side of the slice is given by :

$$x = \sqrt{400 - h^2} \text{ metres.}$$

(ii) Show that the area A of the slice described above is given by :

$$A = \frac{3\sqrt{3}}{2} (400 - h^2)$$

(iii) Hence calculate the volume of the dome.

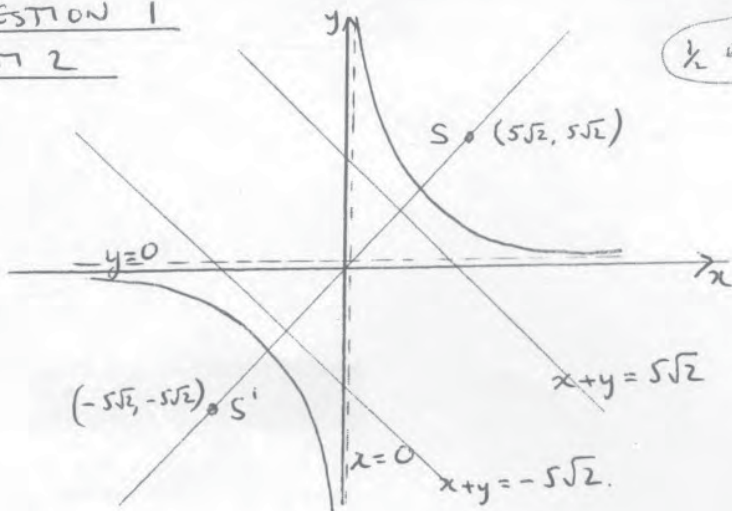
- (b) The point $T\left(ct, \frac{c}{t}\right)$ lies on the rectangular hyperbola $xy = c^2$.
- (i) Derive the equation of the normal at T .
- (ii) The normal at T meets the rectangular hyperbola $x^2 - y^2 = a^2$ at Q and R .
For $0 < t < \frac{a}{c}$ show that T is the midpoint of QR .
- (iii) The rectangular hyperbola $xy = c^2$ is rotated $\frac{\pi}{4}$ radians anti-clockwise about the origin.
- (α) Derive the new Cartesian equation for the hyperbola.
- (β) State the new equations for the asymptotes and directrices.

Question 4

- (a) The region bounded by the x axis, the line $x = \frac{\pi}{4}$ and the curve $y = \tan x$ is rotated about the line $y = -1$.
Find the volume V of revolution.
- (b) On a smooth horizontal table a body with a 2 kg mass is attached to a spring which is pulled to a position x .
The force F Newtons exerted by the spring towards the origin is given by: $F = 26x + 6x^2$ where x metres is the displacement of the mass measured from a fixed point O at time t seconds.
- (i) Draw a force diagram showing all the forces on the body.
- (ii) If the body has a velocity of v m/s and is initially at rest 1 metre to the right of O , show that : $v^2 = 15 - 13x^2 - 2x^3$.
- (iii) Find the displacement when the body next comes to rest.
- (iv) Find the maximum speed of the body during the motion.
- (v) Describe the motion of the body ,showing that the body oscillates between two points.

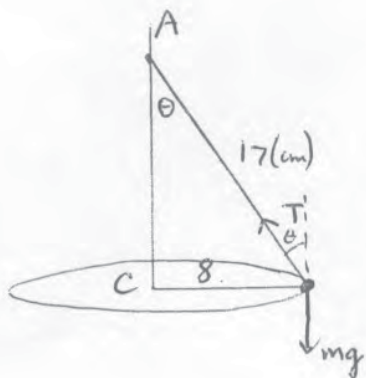
1 a) $xy = c^2$
 $c = 5$

EXT 2 QUESTION 1
 2009 TERM 2



1/2 each

b)



$$\left. \begin{aligned} T \cos \theta &= mg \\ T \sin \theta &= m \omega^2 r \end{aligned} \right\} \text{Dividing } \tan \theta = \frac{\omega^2 r}{g} = \frac{8}{15}$$

$$\omega = \sqrt{\frac{8g}{15r}} \text{ rad/sec. } r = 0.08 \text{ (m)}$$

$$g = 10$$

$$\therefore \omega = \sqrt{\frac{80}{1.2}} = \underline{\underline{8.2 \text{ rad/sec (2 S.F.)}}}$$

④ -1 off each error

c) i) $x = v - \tan^{-1} v$

$$\frac{dx}{dv} = 1 - \frac{1}{1+v^2} = \frac{v^2}{1+v^2} \quad (1)$$

$$\frac{dv}{dx} = \frac{1+v^2}{v^2}$$

$$\ddot{x} = v \frac{dv}{dx} = \frac{1+v^2}{v} = \frac{1}{v} + v \quad (1)$$

ii) $\frac{dv}{dt} = \frac{1}{v} + v = \frac{1+v^2}{v}$

$$\therefore \frac{dt}{dv} = \frac{v}{1+v^2}$$

$$t = \frac{1}{2} \ln(1+v^2) + c$$

When $t=0, v=0 \therefore c=0$

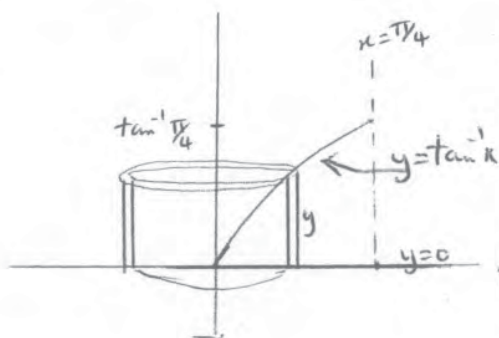
$$t = \frac{1}{2} \ln(1+v^2)$$

$$1+v^2 = e^{2t}$$

$$v^2 = e^{2t} - 1$$

$$v = \sqrt{e^{2t} - 1} \quad (\text{moves to right})$$

d)



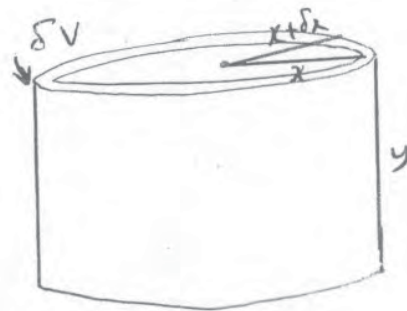
$$\delta V = (\pi(x+\delta x)^2 - \pi x^2) \delta y$$

$$\delta V = 2\pi x y \delta x \quad (1)$$

(neglecting second order terms).

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi/4} 2\pi x y \delta x$$

$$= 2\pi \int_0^{\pi/4} x y \, dx = 2\pi \int_0^{\pi/4} x \tan^{-1} x \, dx \quad (1)$$



$$V = 2\pi \left[\frac{x^2 \tan^{-1} x}{2} \right]_0^{\pi/4} - 2\pi \int_0^{\pi/4} \frac{x^2}{2(1+x^2)} \, dx$$

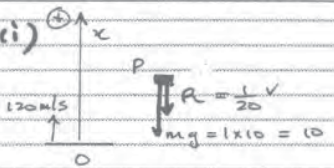
$$= 2\pi \left[\frac{x^2 \tan^{-1} x}{2} \right]_0^{\pi/4} - \pi \left[x - \tan^{-1} x \right]_0^{\pi/4} \quad (1)$$

$$= 2\pi \cdot \frac{\pi^2}{32} \tan^{-1} \frac{\pi}{4} - \frac{\pi^2}{4} + \pi \tan^{-1} \frac{\pi}{4} = \frac{\pi(\pi^2 + 16) \tan^{-1} \frac{\pi}{4}}{16} - \frac{\pi^2}{4} \text{ units}^3 \quad (1)$$

$$= 0.914 \frac{\text{m}^3}{\text{cm}^3}$$

- 0.914 (2 d.p.)

MATH EXT 2 SOLUTIONS TERM 2, 2009

MATHEMATICS Extension 2: Question 2		
Suggested Solutions	Marks	Marker's Comments
Q2(a) (i)  mass = 1 kg $R = \frac{1}{20}v$ $mg = 1 \times 10 = 10$ $\boxed{1}$		$\frac{1}{2}$ For $\uparrow$ and $\downarrow R$ $\frac{1}{2}$ For $R = \frac{1}{20}v$
(ii) Force equation $F = m\ddot{x} = -mg - R$, $m = 1$ (Newtons 2nd) $\therefore 1 \cdot \ddot{x} = -1 \times 10 - \frac{1}{20}v$ $\therefore \ddot{x} = -10 - \frac{1}{20}v$ $\boxed{1}$	2 nd	1 For evidence $-\frac{1}{2}$ if $-\frac{mv}{20}$ and $\frac{1}{2} m$
(iii) $\ddot{x} = v \frac{dv}{dx} = -\frac{1}{20}(200 + v)$ when $t = T$, $x = H$ and $v = 0$ $t = 0$, $x = 0$, $v = 120$ $\int_{120}^0 \frac{v dv}{200 + v} = -\frac{1}{20} \int_0^H dx$ $\int_{120}^0 \left(1 - \frac{200}{200 + v}\right) dv = -\frac{H}{20}$ $\left[v - 200 \ln 200 + v \right]_{120}^0 = -\frac{H}{20}$ as $v > 0$, $200 + v > 0$ $\therefore \left[(0 - 200 \ln 200) - (120 - 200 \ln 320) \right] = -\frac{H}{20}$ $\therefore H = 20 \left[120 + 200 \ln \frac{200}{320} \right] = 20 \left[120 + 200 \ln \frac{5}{8} \right]$ $= 519.985483...$ $\therefore \text{max height} = 520 \text{ m}$ $\boxed{3}$	$\frac{1}{2}$ $C = 20[120 - 200 \ln \frac{320}{200}]$ $V = 20[120 - V + 200 \ln \frac{(200+V)}{320}]$ $\frac{1}{2}$ show evidence	
(iv) $\ddot{x} = \frac{dv}{dt} = -\frac{1}{20}(200 + v)$ From 3U: $v = -200 + Ae^{-\frac{t}{20}}$ $t = 0$, $v = 120$ gives $A = 320$ $v = -200 + 320e^{-\frac{t}{20}}$ so $t = T$, $v = 0$, $0 = -200 + 320e^{-\frac{T}{20}}$ will give $T = 20 \ln \frac{320}{200} = 20 \ln \frac{8}{5} = 9.400072583...$ $\therefore \text{Time} \approx 9.4 \text{ sec}$ $\boxed{2}$		from "Further G+D" DE
OR $\int_{120}^0 \frac{dv}{200 + v} = -\frac{1}{20} \int_0^T dt$ $\ln(200 + v) \Big _{120}^0 = -\frac{T}{20}$ $T = -20 \ln \frac{200}{320} = 20 \ln \frac{320}{200} = \text{etc}$		

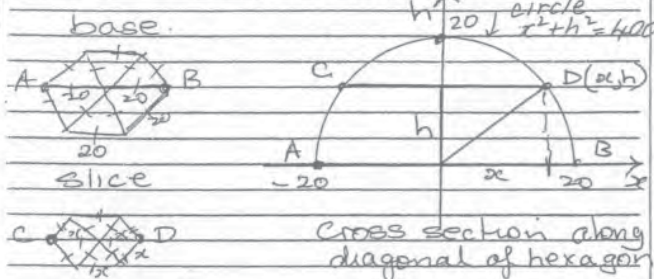
MATHEMATICS Extension 2: Question 2		
Suggested Solutions	Marks	Marker's Comments
Q2(b) (i) $\uparrow x$ $R = \frac{1}{20}v$ $mg = 10$ $\boxed{1}$		1 For direction $\downarrow$ and $\uparrow R$
(ii) $m\ddot{x} = mg - R$ (Newtons 2nd) $\therefore 1 \cdot \ddot{x} = 10 - \frac{1}{20}v$, $m = 1$ $\therefore \ddot{x} = 10 - \frac{1}{20}v$ For terminal velocity $\lim_{t \rightarrow \infty} \ddot{x} = 0$ so $10 - \frac{v_T}{20} = 0$, $v_T = 200$ $\therefore$ terminal velocity is 200 m/s $\boxed{1}$		early $v = 200(1 - e^{-\frac{t}{20}})$ $t \rightarrow \infty$, $v \rightarrow 200$
(iii) $\ddot{x} = \frac{dv}{dt} = \frac{1}{20}(200 - v)$ and $0 \leq v < 200$ From 3U: $v = 200 + Be^{-\frac{t}{20}}$ $t = 0$, $v = 0$ gives $B = -200$ $\therefore v = 200(1 - e^{-\frac{t}{20}})$ so when $t = T \approx 9.4$ $v = 200(1 - e^{-0.47}) = 74.99954624...$ $v = 75$ (3SF) $\therefore$ velocity is 75 m/s OR $\int_0^T \frac{dv}{200 - v} = \frac{1}{20} \int_0^T dt$ when $t = 0$, $v = 0$ $-\ln 200 - v \Big _0^T = \frac{T}{20}$ $-\ln \frac{(200 - v)}{200} = \frac{T}{20} = \frac{9.4...}{20} = 0.47...$ $\frac{200 - v}{200} = e^{-0.47}$ $v = 200(1 - e^{-0.47}) = \text{etc}$ but $200 - v > 0$ and $0 \leq v < 200$, $200 - v = 200 - v$ $\boxed{2}$		$T = 200 \ln \frac{8}{5}$ 1 For $v = 200(1 - e^{-\frac{t}{20}})$ 1 For velocity is 75 m/s (3SF) * big problems if have $v - 200$!!!
(iv) From (iii) $v = \frac{dx}{dt} = 200(1 - e^{-\frac{t}{20}})$ $\int_0^T dx = 200 \int_0^T (1 - e^{-\frac{t}{20}}) dt$ $x \Big _0^T = x = 200 \left[t + 20e^{-\frac{t}{20}} \right]_0^T$ $= 200 \left[(T + 20e^{-\frac{T}{20}}) - (0 + 20) \right]$ $= 200 \left[9.4 + 12.5000... - 20 \right] = 380.0000731...$ $x = 380$ (3SF) OR $\frac{v dv}{dx} = \frac{1}{20}(200 - v)$ $t = 0$, $x = 0$, $v = 0$ $\int_0^T \frac{v dv}{200 - v} = \frac{1}{20} \int_0^T dx$ $t = T$, $x = x$, $v = 75$ $\int_0^{75} \left(-1 + \frac{200}{200 - v} \right) dv = \frac{x}{20}$ $\left[-v - 200 \ln(200 - v) \right]_0^{75} = \frac{x}{20}$ will give $x = -20 \left[75 + 200 \ln \frac{8}{5} \right]$ $\boxed{4}$		$t = 0$, $x = 0$ $t = T$, $x = x$ $T = 200 \ln 1.6 \div 9.4$ $-v + 200$ $\frac{-1}{v - 200}$ $\frac{0 + 200}{0 + 200}$ $\therefore$ distance above the ground $= H - 280$ $= 140 \text{ m}$ (3SF) distance

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

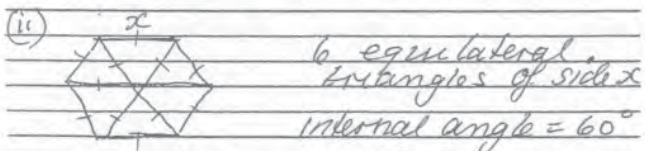
Marker's Comments



By Pythagoras or equation of circle

$$x^2 + h^2 = 400$$

$$x = \sqrt{400 - h^2}$$



Area is $6 \times \frac{1}{2} x^2 \sin 60$

$$A = 3 \sqrt{3} (400 - h^2) \times \frac{1}{2}$$

$$A = \frac{3\sqrt{3}}{2} (400 - h^2)$$

(iii) HENCE

$$\delta V = A \delta h$$

$$V = \lim_{\delta h \rightarrow 0} \sum_{h=0}^{20} \frac{3\sqrt{3}}{2} (400 - h^2) \delta h$$

$$= \int_0^{20} \frac{3\sqrt{3}}{2} (400 - h^2) dh$$

$$= \frac{3\sqrt{3}}{2} \left[400h - \frac{h^3}{3} \right]_0^{20}$$

$$= 8000\sqrt{3}$$

Volume is $8000\sqrt{3} \text{ m}^3$

Must show how hexagons and semi circles are related.

Must show a triangle area formula.

As it is HENCE

① must show use of δV and $\lim_{\delta h \rightarrow 0}$

① Integral

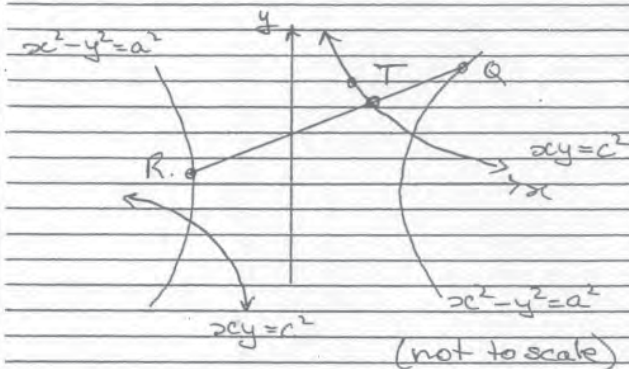
① correct answer.

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments



(i) $xy = c^2$

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

at $T(ct, ct)$ $m_T = -\frac{c/t}{ct} = -\frac{1}{t^2}$

equation of Normal

$$y - \frac{c}{t} = -t^2(x - ct)$$

$$y = t^2x + c(1 - t^3)$$

(ii) Intersection with $x^2 - y^2 = a^2$

$$x^2 - (t^2x + c(\frac{1}{t} - t^3))^2 = a^2$$

$$(1 - t^4)x^2 - 2ct(1 - t^4)x - c^2(\frac{1}{t} - t^3)^2 - a^2 = 0$$

midpoint of Q & R = $-\frac{b}{2a}$

$$= \frac{2ct(1 - t^4)}{2(1 - t^4)}$$

$$= ct$$

as y is on normal

$$y = t^2x + \frac{c}{t} - ct^3$$

$$= \frac{c}{t}$$

$\therefore$ midpoint

① gradient of tangent

① equation of normal (any form)

① equation in t for point of intersection

① x value

① y value

T2 2009

42

MATHEMATICS: Question 4

Suggested Solutions

Marks

Marker's Comments

a) Disc Method

$$\Delta V \doteq \pi[(1 + \tan x)^2 - 1^2] \Delta x = \pi(2 \tan x + \tan^2 x) \Delta x$$

$$V = \pi \int_0^{\frac{\pi}{4}} 2 \tan x + \tan^2 x \, dx$$

$$= \pi [-2 \ln(\cos x) + \tan x - x]_0^{\frac{\pi}{4}}$$

$$\text{Vol} = \pi \left[1 - \frac{\pi}{4} + \ln 2 \right] \text{unit}^3$$

Shell Method

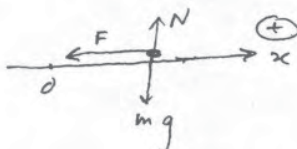
$$\Delta V \doteq 2\pi(y+1)\left(\frac{\pi}{4} - x\right) \Delta y$$

$$V = 2\pi \int_0^1 (y+1) \left[\frac{\pi}{4} - \tan^{-1} y \right] dy$$

$$V = \frac{3\pi}{4} - (\pi^2 - \pi - \pi \ln 2)$$

$$\text{Vol} = \pi \left[1 - \frac{\pi}{4} + \ln 2 \right] \text{unit}^3$$

b i)



$$\text{ii) } -m\ddot{x} = 26x + 6\dot{x} \quad (m=2)$$

$$\text{ie. } v \frac{dv}{dx} = -13x - 3\dot{x}$$

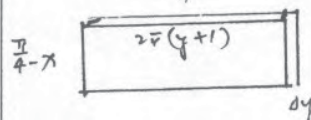
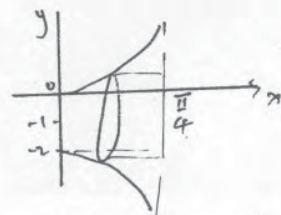
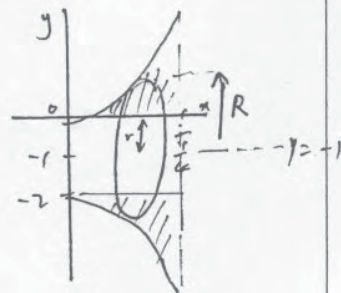
$$\int v \, dv = \int -13x - 3\dot{x}^2 \, dx$$

$$v^2 = 15 - 13x^2 - 2x^3$$

$$\text{iii) } v^2 = (1-x)(2x^2 + 15x + 15)$$

$$x \neq 1 \quad x = \frac{-15 \pm \sqrt{105}}{4} \quad (-1.19, -6.31)$$

$$\text{next stop at } x = \frac{-15 + \sqrt{105}}{4}$$



$$\frac{1}{2} m v$$

$$\frac{1}{2} m g$$

1 m F towards origin
(2 forces no marks)

many fudging answers

MATHEMATICS: Question 4

Suggested Solutions	Marks	Marker's Comments
biv) For max v $v \frac{dv}{dx} = 0$ $-13x - 3x^2 = 0$ $-x(13 + 3x) = 0$ $\therefore x = 0 \text{ or } -\frac{13}{3}$ When $x = -\frac{13}{3}$ $v^2 = -66\frac{10}{27} \therefore x \neq -\frac{13}{3}$ $x = 0$ $v = \pm\sqrt{15}$ $\therefore$ max speed is $\sqrt{15}$ m/s v) At $x=1$, $v=0$ $\ddot{x} = -16$ m/s² $x=0$, $v=\sqrt{15}$ $\ddot{x}=0$ $x = \frac{-15 + \sqrt{105}}{4}$ $\ddot{x} = 11.2$ (m/s²) Initially starts at rest from $x=1$ m moving left ($\ddot{x} = -16$) $\frac{1}{2}$ m increasing speed reaching max of $\sqrt{15}$ m/s at $\frac{1}{2}$ m origin, slow down & stops at $x = \frac{-15 + \sqrt{105}}{4}$ $\frac{1}{2}$ m Turn around moving right ($\ddot{x} = 11.2 > 0$) $\frac{1}{2}$ m stops at $x=1$. oscillates between $\frac{-15 + \sqrt{105}}{4} \approx -1.19$ and $x=1$ $\frac{1}{2}$ m vi) $T = 2 \int_{\frac{-15 + \sqrt{105}}{4}}^1 \frac{dx}{\sqrt{15 - 13x^2 - 2x^3}}$ $\frac{dx}{dt} = \sqrt{15 - 13x^2 - 2x^3}$ for $t_2 \leq t \leq T$	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ m $\frac{1}{2}$ m $\frac{1}{2}$ m $\frac{1}{2}$ m 1 m	many students forgot to justify $x \neq -\frac{13}{3}$ students got the end pts wrong (from previous parts) max $1\frac{1}{2}$ m only $\frac{1}{2}$ m for each mistake badly done